

MLT (Assignment)

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- ECE 4077
- Patch 2.

1) Probability theory fed representing uncertainty Bayesian classifier.

class:

C_1 : buys - computer = "Yes".

C_2 : buys - computer = "No"

Data to be classifier:

age ≤ 30 ;

income = medium;

student = Yes

credit = rating = fair.

$$P(C|u) = \frac{P(u|C) \cdot P(C)}{P(u)}$$

$$P(C|u) = \frac{P(u_1|C_1) \cdot P(C_1) + P(u_2|C_2) \cdot P(C_2)}{P(u_1|C_1) \cdot P(C_1) + P(u_2|C_2) \cdot P(C_2)}$$

age	income	student	credit - rating	computer
≤ 30	high	No	fair	No
≤ 30	high	No	excellent	No
31 ... 40	high	No	fair	Yes
> 40	medium	Yes	fair	Yes
> 40	low	Yes	fair	Yes
> 40	low	Yes	excellent	No
31 ... 40	low	Yes	excellent	Yes
≤ 30	medium	No	fair	No
≤ 30	low	Yes	fair	Yes
> 40	medium	Yes	fair	Yes
≤ 30	medium	Yes	excellent	Yes
31 ... 40	medium	No	excellent	Yes
31 ... 40	high	Yes	fair	Yes
31 ... 40	high	No	excellent	No
> 40	medium	No	excellent	No

$$P(G_1) : P(\text{buys - computer} = \text{"Yes"}) = 9/14 = 0.643.$$

$$P(\text{buys - computer} = \text{"No"}) = 5/14 = 0.307.$$

* computer $P(u|c_i)$ for each class

$$P(\text{age} = "<230" | \text{buys} = \text{computer} = \text{"Yes"}) = 2/9 = 0.222.$$

$$P(\text{age} = "<30" | \text{buys} = \text{computer} = \text{"No"}) = 3/5 = 0.6$$

$$P(\text{income} = \text{"medium"} | \text{buys} = \text{computer} = \text{"Yes"}) = 4/9 = 0.444.$$

$$P(\text{income} = \text{"medium"} | \text{buys} = \text{computer} = \text{"No"}) = 2/5 = 0.4.$$

$$P(\text{student} = \text{"Yes"} | \text{buys} = \text{computer} = \text{"Yes"}) = 6/9 = 0.667$$

$$P(\text{student} = \text{"Yes"} | \text{buys} = \text{computer} = \text{"No"}) = 1/5 = 0.2.$$

$$P(\text{credit-rating} = \text{"fair"} | \text{buys} = \text{computer} = \text{"Yes"}) = 6/9 = 0.667.$$

$$P(\text{credit-rating} = \text{"fair"} | \text{buys} = \text{computer} = \text{"No"}) = 3/5 = 0.4.$$

* $u = (\text{age} < 230, \text{income} = \text{medium}, \text{student} = \text{Yes}, \text{credit-rating} = \text{fair})$.

$$P(u|c_i) = P(u | \text{buys} = \text{computer} = \text{"Yes"}) = 0.222 \times 0.444 \times 0.667 \times 0.667$$

$$= 0.044.$$

$$= 0.6 \times 0.4 \times 0.2 \times 0.4.$$

$$= 0.019.$$

$$P(u|cp) * P(c_i) = P(u | \text{buys} = \text{computer} = \text{"Yes"}) * P(\text{buys} = \text{computer} = \text{"Yes"})$$

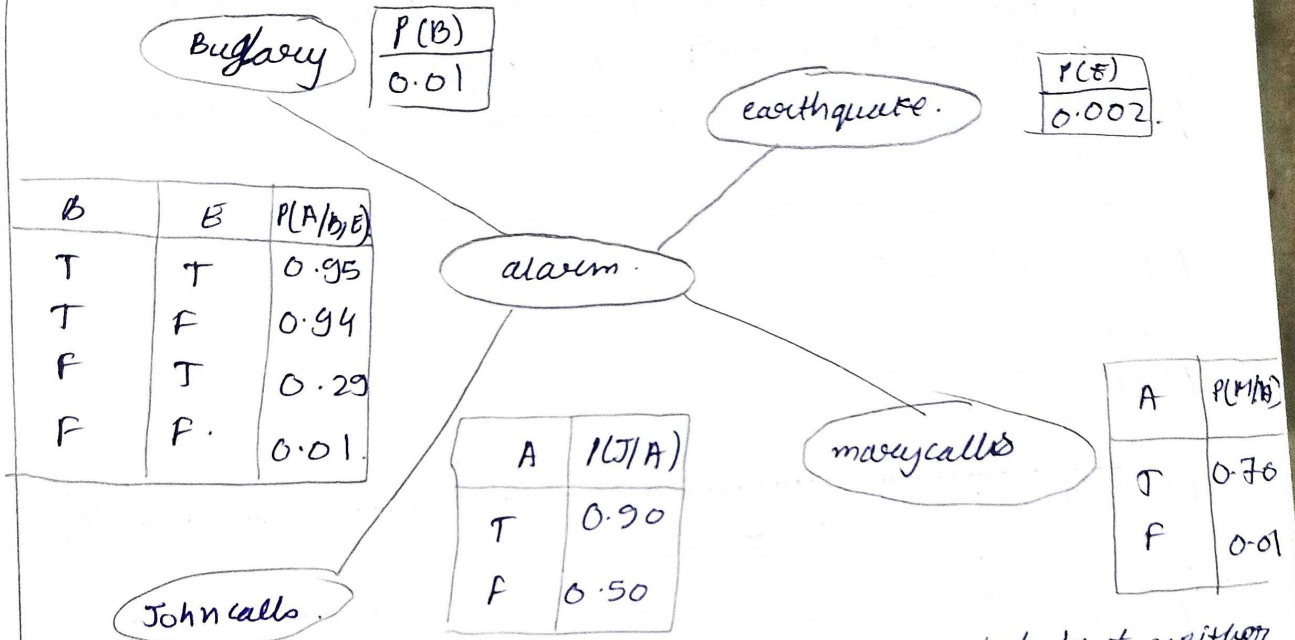
$$= 0.028.$$

$$P(u | \text{buys} = \text{computer} = \text{"No"}) * P(\text{buys} = \text{computer} = \text{"No"}).$$

$$= 0.007.$$

Therefore, u belongs to class ("buys-computer, Yes").

Bayesian belief network.



1) what is the probability that the alarm has sounded but neither a burglary nor an earthquake has occurred, and both John and Mary call

Sol:-

$$P(J \wedge M \wedge A \wedge \neg B \wedge \neg E) = P(J|A) \cdot P(M|A) \cdot P(A|\neg B, \neg E) \cdot P(\neg B) \cdot P(\neg E)$$

$$= 0.90 \times 0.70 \times 0.001 \times 0.999 \times 0.998$$

$$= 0.00062$$

2). what is the probability that the neither John nor Mary calls

Sol:-

$$P(\neg J, \neg M)$$

$$P(\neg J, \neg M) = P(\neg J, \neg M) \cdot P(A) + P(\neg J, \neg M | \neg A) \cdot P(\neg A)$$

$$P(\neg J, \neg M | A) = P(\neg J|A) \cdot P(\neg M|A)$$

$$= 0.10 \times 0.50$$

$$= 0.05$$

$$P(\neg J, \neg M | \neg A) = P(\neg J|\neg A) \cdot P(\neg M|\neg A)$$

$$= 0.95 \times 0.99$$

$$= 0.9405$$

$$P(A) = P(A|B)P(E)$$

$$P(A|B, E) \cdot P(B) \cdot P(E) + P(A|B, \neg E) \cdot P(B) \cdot P(\neg E) + P(A|\neg B, E) \cdot P(\neg B) \cdot P(E) + P(A|\neg B, \neg E) \cdot P(\neg B) \cdot P(\neg E)$$

$$P(A) = (0.95 \times 0.001 \times 0.002) + (0.94 \times 0.001 \times 0.998) + (0.29 \times 0.999 \times 0.002) + (0.001 \times 0.999 \times 0.998)$$

$$= 0.0000019 + 0.00093812 + 0.00057942 + 0.000997002$$

$$P(A) = 0.002516442$$

Thus

$$P(\neg A) = 1 - 0.002516$$

$$= 0.997484$$

$$P(J, M) = (0.03)(0.002516) + (0.9705)(0.997484)$$

$$= 0.00007548 + 0.937449$$

$$P(J, M) = 0.9376$$

3) What is the probability that the alarm ring given that there is a burglary but no earthquake?

$$\text{Sol } P(A|B, \neg E) = 0.97$$

4) What is the probability the masonry calls given that the alarm rings

$$\text{Sol: } P(M|A) = 0.70$$

5) What is the probability that the alarm rings given that there is no burglary & no earthquake?

$$\text{Sol: } P(A|\neg B, \neg E) = 0.001$$

6) What is the probability that John calls but Mary does not call?

$$\text{Sol: } P(J \wedge \neg M)$$

$$P(J \wedge \neg M) = P(J, \neg M|A) \cdot P(A) + P(J, \neg M|\neg A) \cdot P(\neg A)$$

$$P(J, \neg M|A) = P(J|A) \cdot P(\neg M|A)$$

$$P(J, \neg M|\neg A) = P(J|\neg A) \cdot P(\neg M|\neg A)$$

$$= 0.05 \times 0.99$$

$$= 0.0495$$

$$P(J|M) = (0.27 \times 0.00251644) + (0.0495 \times 0.99718556) \\ = 0.0067$$

$$0.143 + 0.0493 = 0.1923$$

3) Given the hypothesis space with 4 hypothesis h_1, h_2, h_3 & h_4 determine if the patient is diagnosed as covid positive or covid negative using optimal classifier.

$P(h_i T)$	$P(\text{covid positive} h_i)$	$P(\text{covid negative} h_i)$
0.3	0	1
0.1	1	0
0.2	1	0
0.1	1	0

h_{map} choose h_1 which has the maximum probability value 0.3 as the selection and gives the result that the patient is covid negative. But Bayes optimal classifier combines the predictions of h_2, h_3 and h_4 which is 0.4 and gives the result that the patient is covid positive.

$$\sum_{h_i \in H} P(\text{covid negative} | h_i) \cdot P(h_i | T) = 0.3 \times 1 = 0.3$$

$$\sum_{h_i \in H} P(\text{covid positive} | h_i) \cdot P(h_i | T) = 0.1 \times 1 + 0.2 \times 1 + 0.1 \times 1 = 0.4$$

Therefore, $\text{argmax}_{c \in \{\text{covid positive}, \text{covid negative}\}} \sum_{h_i \in H} P(c | h_i) \cdot P(h_i | T) = \text{covid positive}$.

Thus, this algorithm, diagnosis the new instance to be covid positive.